



# SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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## Department of Computer Science and Engineering DevOps Lab - L3

| Lab Assignment No: 10    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| PRN                      | 23070122202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Batch                    | 2023 - 2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Class                    | CSE – B1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Academic Year & Semester | 2026 - 2027, 7th Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Title of Assignment      | Docker Volumes and Data Persistence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation & Theory  | <pre>assignment - 10 &gt; app.py &gt; ... 1  from flask import Flask 2  from datetime import datetime 3  import os 4 5  app = Flask(__name__) 6 7  DATA_DIR = "/app/data" 8  LOG_FILE = os.path.join(DATA_DIR, "greeting_log.txt") 9 10 @app.route("/") 11 def home(): 12     os.makedirs(DATA_DIR, exist_ok=True) 13 14     with open(LOG_FILE, "a") as f: 15         f.write(f"Request received at {datetime.now()}\n") 16 17     with open(LOG_FILE, "r") as f: 18         logs = f.read() 19 20     return f""" 21     &lt;h1&gt;Hello, Docker!&lt;/h1&gt; 22     &lt;h2&gt;Request Log&lt;/h2&gt; 23     &lt;pre&gt;{logs}&lt;/pre&gt; 24     """ 25 26 if __name__ == "__main__": 27     app.run(host="0.0.0.0", port=5000)</pre> <p>Modified app.py</p> |

```

PS E:\DevOps_SEM_7\Assignment - 10> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
my-flask-app         v2                 e647443a25d2       50 seconds ago     200MB
my-flask-app         latest            2aec46790c9e       28 hours ago       200MB
qdrant/qdrant        latest            057ee3a8da76       2 weeks ago        270MB
neo4j                latest            2ae0454ba8         2 weeks ago        1.07GB
jenkins/jenkins      lts-jdk21         f4f65e6cd140       6 weeks ago        801MB
jenkins/jenkins      lts-jdk17         cce1eeb79902       5 months ago       785MB
PS E:\DevOps_SEM_7\Assignment - 10>

```

**Image Rebuilt:** The Flask application was modified to record timestamped requests and was rebuilt as my-flask-app:v2. This updated image is used for testing persistent storage with Docker volumes.

```

PS E:\DevOps_SEM_7\Assignment - 10> docker volume ls
DRIVER      VOLUME NAME
local       bc2ca379fde3781db2f8164afe0554b06c9b1dd4ff55a76228cb5a6172cb9aa5
local       c9f0a53d7008c016c1d8a62894edb5be12025f6d9801e88619fb2ff5091e8150
local       e0e350e99647e6d8022d65f4c1a83e12f5d68fee70ceb1433059390b33b16512
local       jenkins_home
local       my-app-data
local       n8n_data
local       neo4j_data
local       qdrant_data
PS E:\DevOps_SEM_7\Assignment - 10>

```

**Docker Volume Creation:** A named Docker volume my-app-data was created to provide persistent storage for the Flask application. Named volumes allow application data to remain available independently of the container lifecycle.

```

PS E:\DevOps_SEM_7\Assignment - 10> docker run -d --name my-flask-container -p 5000:5000 -v my-app-data:/app/data my-flask-app:v2
45a4d658266b416b539ea3a99fa8dd36096d5db2a61d694700a62efab7fe2baa
PS E:\DevOps_SEM_7\Assignment - 10> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
45a4d658266b   my-flask-app:v2 "python app.py"         9 seconds ago Up 8 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   my-flask-container
PS E:\DevOps_SEM_7\Assignment - 10> docker exec my-flask-container cat /app/data/greeting_log.txt
Request received at 2026-08-21 08:21:49.781244
Request received at 2026-08-21 08:21:56.316058
Request received at 2026-08-21 08:21:57.040075

```

**Running the Container with a Volume:** The previous Assignment 8 container was removed and recreated using the updated my-flask-app:v2 image. The named volume my-app-data was mounted to /app/data, allowing the Flask application to store request logs in persistent Docker storage.

```

PS E:\DevOps_SEM_7\Assignment - 10> docker stop my-flask-container
my-flask-container
PS E:\DevOps_SEM_7\Assignment - 10> docker rm my-flask-container
my-flask-container
PS E:\DevOps_SEM_7\Assignment - 10> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
cc111857475a   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u..." 11 days ago    Exited (143) 10 days ago           myjenkins
9f64d7378874   qdrant/qdrant                       ".entrypoint.sh"         2 weeks ago    Exited (143) 4 days ago           qdrant
d88b7ca7114e   neo4j                                "tini -g -- /startup..." 2 weeks ago    Exited (137) 4 days ago           neo4j
PS E:\DevOps_SEM_7\Assignment - 10> docker volume ls
DRIVER         VOLUME NAME
local          bc2ca379fde3781db2f8164afe0554b06c9b1dd4ff55a76228cb5a6172cb9aa5
local          c9f8a53d7008c016c1d8a62894edb5be12025fed9801e88619fb2ff5091e8150
local          e0e350e99647e6d8022d65f4c1a83e12f5d68fee70ceb1433059390b33b16512
local          jenkins_home
local          my-app-data
local          n8n_data
local          neo4j_data
local          qdrant_data
PS E:\DevOps_SEM_7\Assignment - 10>

```

**Container Removal:** The original Flask container was stopped and removed while the named volume my-app-data was retained. Since the volume has an independent lifecycle from the container, the application data stored in the volume remains available for future containers.

```

PS E:\DevOps_SEM_7\Assignment - 10> docker run -d --name my-flask-container-2 -p 5000:5000 -v my-app-data:/app/data my-flask-app:v2
a5c636f1cd68593cc62801cc9f03f07605043a0eb8202a854fae936ad0bbffa
PS E:\DevOps_SEM_7\Assignment - 10> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
a5c636f1cd68   my-flask-app:v2                     "python app.py"         31 seconds ago Up 30 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  my-flask-container-2
PS E:\DevOps_SEM_7\Assignment - 10>

```

**New Container with Same Volume:** A new container instance was created from my-flask-app:v2 and the existing my-app-data volume was mounted to /app/data. This demonstrates that the volume can be reused independently of the original container.

```

PS E:\DevOps_SEM_7\Assignment - 10> docker exec my-flask-container-2 cat /app/data/greeting_log.txt
Request received at 2026-08-21 08:21:49.781244
Request received at 2026-08-21 08:21:56.316058
Request received at 2026-08-21 08:21:57.840075
Request received at 2026-08-21 08:53:41.687671
Request received at 2026-08-21 08:53:49.684876
Request received at 2026-08-21 08:53:50.989594
Request received at 2026-08-21 08:53:51.655936
PS E:\DevOps_SEM_7\Assignment - 10>

```

**Persistence Verified:** The new container successfully accessed the log entries created by the previous container. Additional requests generated new entries in the same file. This confirms that the data persisted independently of the original container through the Docker named volume my-app-data.